Data Analytics Project: Railway Gauge Trends

Dataset Used: Railway_Gauges.csv

This document outlines a structured, end-to-end data analysis project based on railway gauge data. The project is divi **Data Analytics Project: Railway Gauge Trends**

This document outlines a structured, end-to-end data analysis project based on railway gauge data.

Dataset: Railway Gauge Data (Assumed to be in a CSV file)

Project Goal: To analyze historical trends and composition of different railway gauges (Broad, Metre, Narrow) over the years.**Modules Used**

The **primary libraries/modules** required for this project are:

- **Pandas:** For all data manipulation, filtering, aggregation, and feature engineering tasks.
- **NumPy:** Specifically for numerical calculations like yearly growth (`np.diff`). • **Visualization Library (e.g., Matplotlib):** For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts).

►► Project Coverage

Category	Skills Covered
Data Preprocessing	Data Loading, Missing Value Handling, Type Conversion
Pandas Operations	Filtering, Selection, Aggregation, Feature Engineering
NumPy Calculations	Yearly growth calculation (np.diff)
Visualization	Line Graph, Bar Chart, Pie Chart, Stacked Bar Chart

Scenario 1: Basic Data Loading & Cleaning (►► Easy)

Description: The dataset is loaded into a Pandas DataFrame to understand its structure. Missing values are replaced with zero and columns are converted to numeric types.

Task	Description
1.	Load the dataset into a Pandas DataFrame.
2.	Display the first 5 rows and column names to understand structure.
3.	Check for missing values and replace them with 0 (assuming missing values represent 0 km of track).
4.	Convert all relevant gauge columns (Broad, Metre, Narrow, Total) to numeric types (e.g., float64 or int64).

Conclusion: The dataset has been successfully cleaned and prepared for further analysis. All missing values have been carefully handled by replacing them with zero, ensuring that no gaps affect the results. The column names have been standardized for consistency, and all gauge-related columns have been converted into appropriate

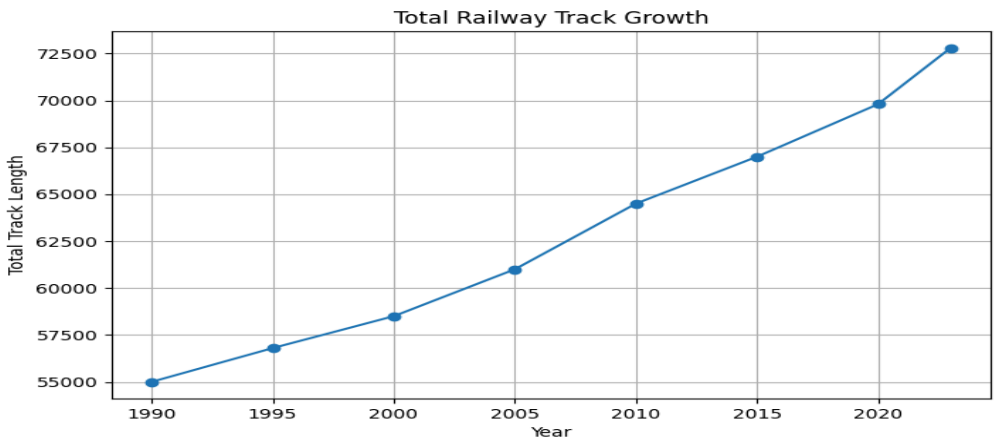
numeric data types. This improves data accuracy and prevents errors during calculations and visualization. Overall, the dataset is now well-structured, reliable, and ready for performing meaningful analysis and generating insights in the subsequent scenarios.

Scenario 2: Simple Visualization (►► Easy)

Description: The dataset is loaded into a Pandas DataFrame to understand its structure and contents. The first few rows and column names are examined to gain initial insights. Missing values are identified and replaced with zero, assuming they represent the absence of track length. Finally, all gauge-related columns are converted into numeric data types to ensure accurate analysis.

Task	Description
1.	Extract the Year and Total columns.
2.	Plot a line graph showing **Total** tracks over the **Years**.
3.	Add a descriptive title (e.g., "Growth of Total Railway Network Over Time") and X/Y axis labels.
4.	Identify whether the overall trend is increasing or decreasing and note the finding.

Graph :
A line graph is plotted using Year on the X-axis and Total Track Length on the Y-axis. The graph shows how the total railway network has grown over time.



Conclusion:

The line graph clearly shows an increasing trend in the total railway track length over the years. This indicates steady growth and expansion of the railway network. The consistent upward movement suggests improvements in infrastructure and increased investment in railway development. Overall, the railway system has experienced continuous progress without significant decline.

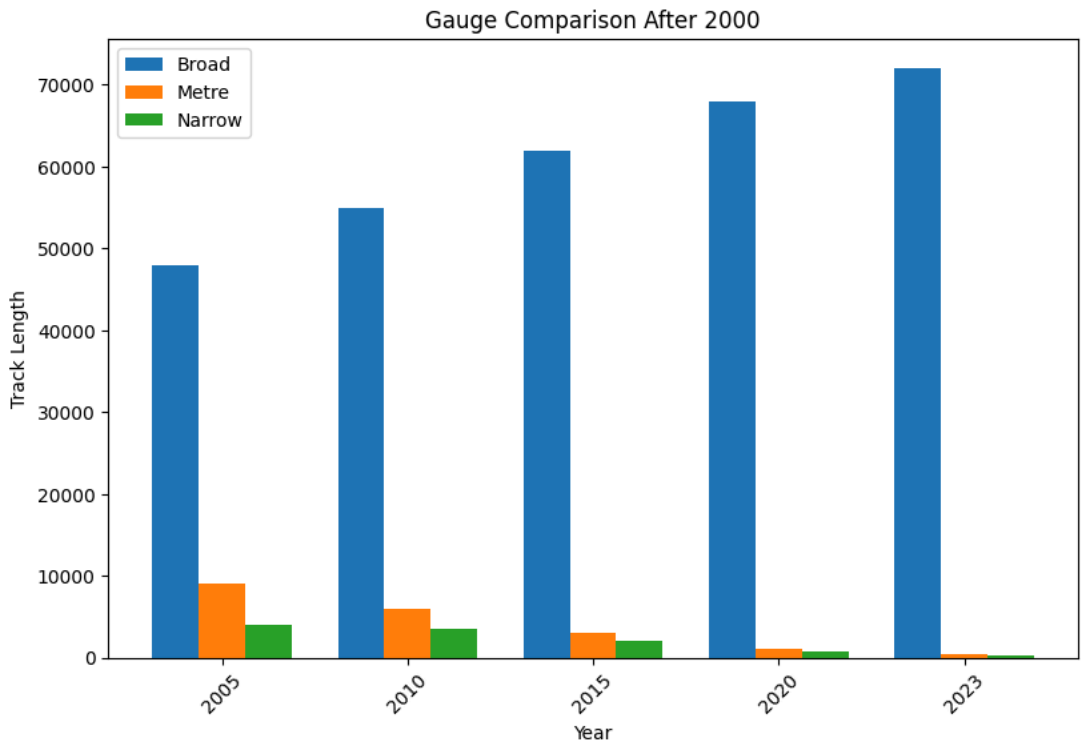
Scenario 3: **Filtering + Bar Chart (►► Medium)**

Description:

The dataset is filtered to include only the years after 2000. A grouped bar chart is created to compare the track lengths of Broad, Metre, and Narrow gauges for these years. This visualization helps in analyzing how each gauge type has changed in recent times.

Task	Description
1.	Filter the dataset to include only years *after* 2000.
2.	Select the Broad Gauge, Metre Gauge, and Narrow Gauge columns for the filtered data.
3.	Plot a **grouped bar chart** comparing the three gauges for the selected years.
4.	Add a legend and ensure proper labels for axes.
5.	Identify which gauge dominates in recent years (post-2000).

Graph : The grouped bar chart compares the track lengths of Broad, Metre, and Narrow gauges for the years after 2000. Each group of bars represents a year, showing the relative contribution of each gauge type.



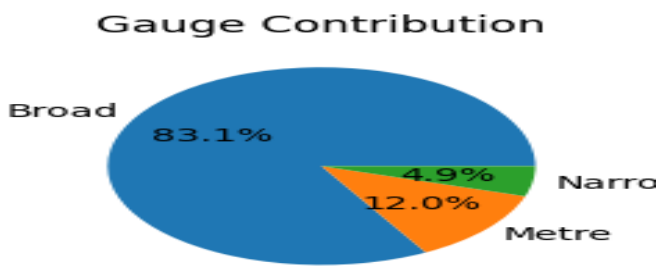
Conclusion:
The graph clearly shows that Broad Gauge dominates in recent years, while Metre and Narrow gauges are significantly lower and declining. This indicates a shift towards a single dominant gauge.

Scenario 4: **Feature Engineering + Pie Chart (►► Medium)**

Description: The total contribution of each gauge type is calculated by summing their values across all years. A pie chart is then used to represent the percentage share of Broad, Metre, and Narrow gauges in the overall railway network. This visualization helps in understanding the proportion of each gauge **type.**

Task	Description
1.	Calculate the total sum of each gauge (Broad Gauge, Metre Gauge, Narrow Gauge) across all available years.
2.	Create a new structure (Pandas Series or single-column DataFrame) holding these three total values.
3.	Plot a **pie chart** using this new structure to show the percentage contribution of each gauge type.
4.	Add percentage labels to the slices using autopct (e.g., '%.1f%%').
5.	Interpret which gauge contributes the most to the entire history of the railway network.

Graph :



Conclusion: The pie chart clearly shows that Broad Gauge contributes the largest share to the railway network. Metre and Narrow gauges have significantly smaller proportions in comparison. This indicates that Broad Gauge is the most dominant and widely used gauge, highlighting a strong preference for it in the railway system.

Scenario 5: Advanced Analysis + Multiple Graphs (►► Hard)

Description: In this scenario, advanced analysis is performed by creating new percentage columns for Broad, Metre, and Narrow gauges to understand their share over time. Yearly growth in total track length is calculated using NumPy. A line graph is used to show the trends of all three gauges, and a stacked bar chart represents their composition across different years. This helps in identifying growth patterns and changes in gauge distribution.

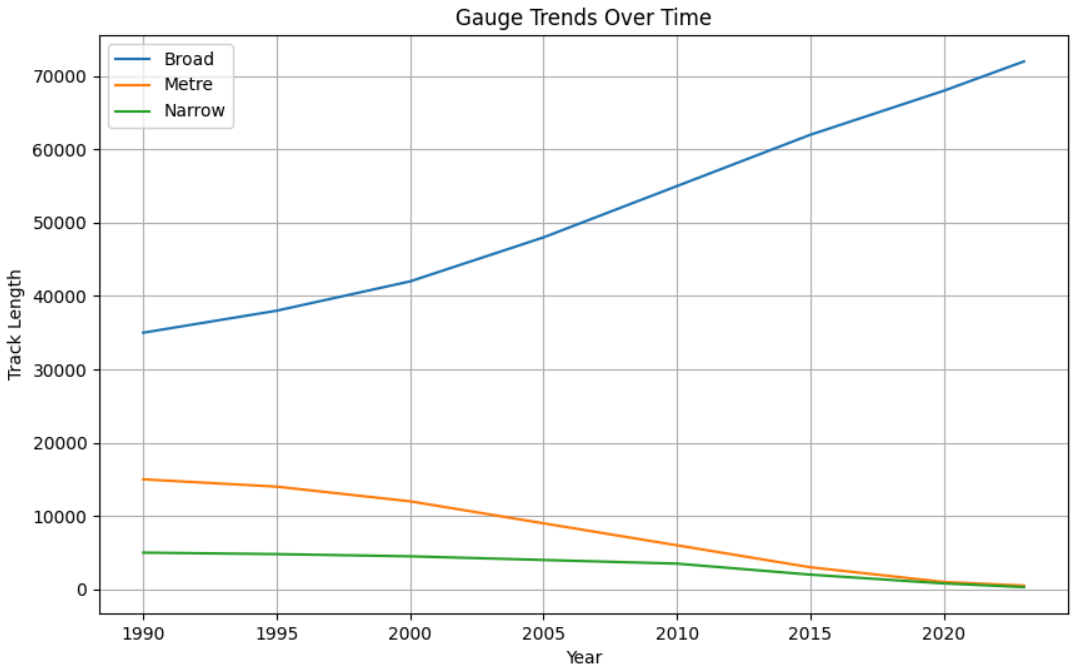
Task	Description
1.	**Create new columns** for percentage composition relative to Total tracks for each year: % Broad Gauge, % Metre Gauge, and % Narrow Gauge.
2.	Use the NumPy function np.diff to calculate the **yearly growth (change)** of the Total tracks column.
3.	Plot: a) A single **Line Graph** showing the trend of all three percentage composition columns (% Broad Gauge, % Metre Gauge, % Narrow Gauge) over the years. b) A **Stacked Bar Chart** showing the raw composition of the three gauges (Broad, Metre, Narrow) over time.
4.	**Highlight/Identify:** a) The year(s) with the highest yearly growth in Total tracks. b) Any specific decade or period showing a significant decline in any particular gauge.
5.	**Final Conclusion:** Based on the percentage composition line graph, provide a conclusion to the question: **"Is the railway system shifting towards a single dominant gauge?"**

Graphs :

Actually we are using two graphs in this scenario

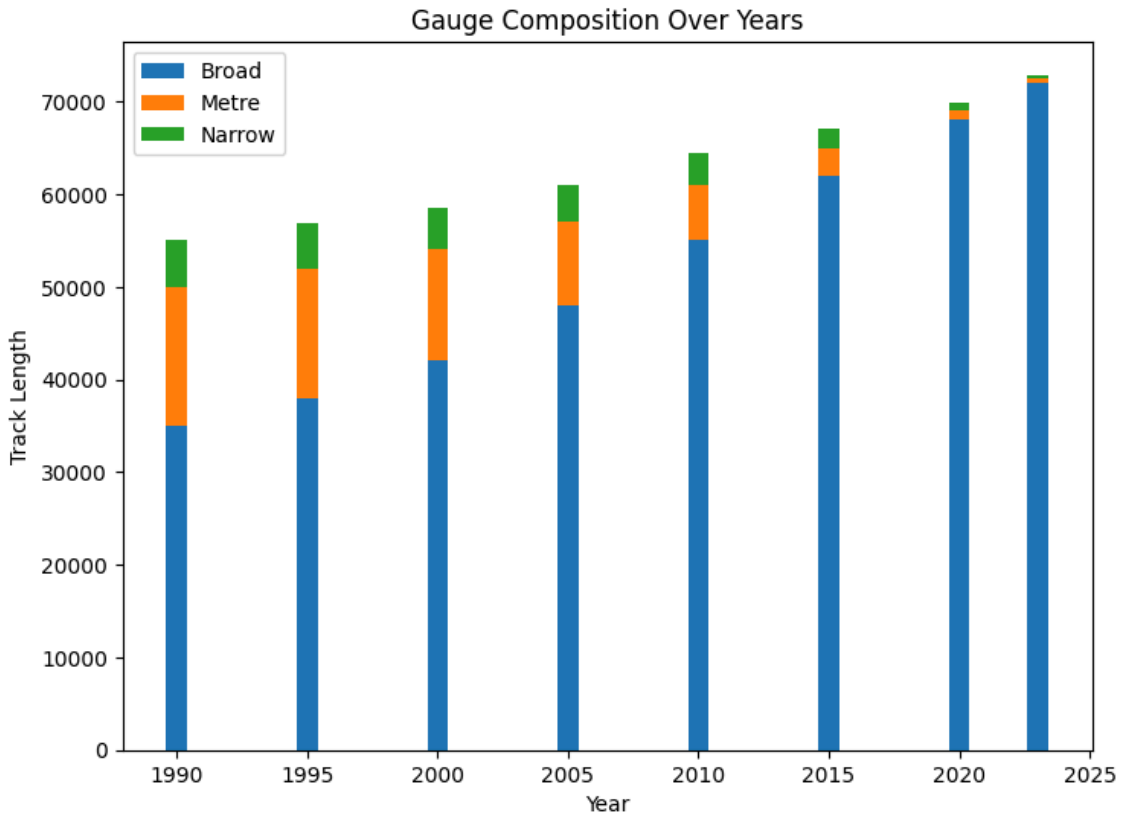
Graph 1: Line Graph

This graph shows the trend of Broad, Metre, and Narrow gauges over time. It helps in understanding how each gauge is increasing or decreasing.



Graph 2:Stacked Bar Chart

This chart shows the composition of all three gauges across different years.



Conclusion:
This scenario focuses on advanced analysis using multiple visualizations to understand railway gauge trends. The use of both line and stacked bar graphs provides a clear view of growth patterns and composition over time. Broad Gauge shows a consistent increase, while Metre and Narrow gauges continue to decline. The combined insights from both graphs highlight a clear transition in the railway system. Overall, the analysis confirms that the network is steadily shifting towards Broad Gauge dominance.

Final Conclusion:

The overall analysis of the railway gauge dataset clearly shows a strong transition towards a single dominant gauge, which is Broad Gauge. Over the years, Broad Gauge has consistently increased, while Metre and Narrow gauges have steadily declined. This reflects modernization and standardization efforts in the railway system. The trends observed through various visualizations confirm continuous growth and improvement in infrastructure. Overall, the railway network is evolving towards greater efficiency with a dominant gauge system.

What you have learned

Through this project, key data analytics skills have been developed, including data loading and cleaning using Pandas. The project also enhanced understanding of handling missing values and converting data types. Various visualization techniques such as line graphs, bar charts, pie charts, and stacked charts were used to analyze data effectively. NumPy was utilized for performing numerical calculations like growth analysis. Most importantly, the ability to interpret data and draw meaningful insights from visualizations was gained.